# Yixuan Xie

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## Education

**University of Illinois at Urbana-Champaign** *08/2026 -- Present*
M.S. in Computer Science -- Advisor: Prof. Jiawei Han *GPA: 4.0/4.0*

**University of Illinois Urbana-Champaign** *08/2022 -- 05/2026*
B.S. in Computer Science & Statistics -- Advisor: Prof. Jiawei Han *GPA: 3.9/4.0*

## Publications

† indicates equal contribution.

**MemComposer: Learning to Assemble Memories for LLM Agents via Reinforcement Learning**
**Yixuan Xie**, Xueqiang Xu, Xinyi Fan, Xinxi Lyu, Tao Feng, Xing Fan, Chenlei Guo, Jie Hao, Shaowen Wang, Jiashuo Sun, Jiawei Han.
In submission to ARR Oct, 2026 [arXiv]

**CaseMem: Closing the Loop from Agent Outcomes to Memory Reuse**
Xinyi Fan, **Yixuan Xie**, Xing Fan, Chenlei Guo, Jie Hao, Jiawei Han.
In submission to ICLR, 2027

**TaSR-RAG: Taxonomy-guided Structured Reasoning for Retrieval-Augmented Generation**
Jiashuo Sun†, **Yixuan Xie**†, Jimeng Shi, Shaowen Wang, Jiawei Han.
In submission to ARR Oct, 2026 [arXiv] [Code]

**Retrieval is Cheap, Show Me the Code: Executable Multi-Hop Reasoning for RAG**
Jiashuo Sun†, Jimeng Shi†, **Yixuan Xie**, Saizhuo Wang, Jash Rajesh Parekh, Pengcheng Jiang, Zhiyi Shi, Jiajun Fan, Qinglong Zheng, Peiran Li, Shaowen Wang, Ge Liu, Jiawei Han.
In submission to ICLR, 2027 [arXiv] [ProjectPage] [Code]

**MultiCube-RAG: Training-Free Multi-Hop RAG by Reasoning over Structured Knowledge Cubes**
Jimeng Shi, Wei Hu, Runchu Tian, Bowen Jin, Wonbin Kweon, SeongKu Kang, Yunfan Kang, Dingqi Ye, Sizhe Zhou, Jiashuo Sun, **Yixuan Xie**, Shaowen Wang, Jiawei Han.
In submission to AAAI, 2027 [arXiv]

## Experience

**Amazon-Illinois AICE** | *Research Assistant* *05/2026 - Present*
Agent Memory Systems *Champaign, IL*

- Proposed **MemComposer**, an RL-trained memory composer that selects and orders memories for LLM agents, outperforming MemRL and Memento on $\tau^2$-bench (+5.9/+4.6 pp avg. task success) with up to 17% fewer memory tokens.
- Introduced **CaseMem**, an outcome-adaptive framework that updates reuse preference based on downstream task's result.

**UIUC Data Mining Group** | *Undergrad Researcher* *08/2025 - 05/2026*
Retrieval-Augmented Generation & RL -- Advised by Prof. Jiawei Han *Champaign, IL*

- Built PyRAG, reformulated multi-hop RAG as executable Python programs with compiler-grounded self-repair.
- Proposed **TaSR-RAG**, a taxonomy-guided structured reasoning framework for multi-hop QA; +14% over baselines.

**Matsuo-Iwasawa Lab** | *Research Assistant* *07/2025 - 12/2025*
LLM Post-Training -- Mentored by Dr. Qian Niu *Remote*

- Fine-tuned Qwen3 series models with SFT and GRPO, enhanced reasoning on medical QA tasks.

**IBM-Illinois Discovery Accelerator Institute** | *Research Assistant* *05/2025 - 08/2025*
Open-Schema Information Extraction -- Advised by Prof. Jiawei Han *Champaign, IL*

- Benchmarked training-free IE methods and LLM in-context learning across DocEE, MAVE, and NERRE.
- Conducted cross-method error analysis to compare robustness across open-schema IE settings.

**DataLynn, Inc.** | *Machine Learning Engineer Intern* *05/2024 - 08/2024*
Retrieval-Augmented Generation *Long Island City, NY*

- Worked on RAG pipelines for AI agents in customer support and interview prep.

## Selected Projects

**Token-Perplexity Signals for Stable RL Post-Training** | *Team Lead* *09/2025 - 12/2025*
Machine Learning System Course Project -- Mentored by Prof. Fan Lai [Tech Report]

- Investigated gradient-based and token-level diagnostic signals for RLHF dynamics, implementing pipeline instrumentation to capture objective-specific gradients and prefix perplexity during GRPO training.
- Systematically analyzed gradient similarity and perplexity distributions across model scales, revealing limited sensitivity of gradient conflict and substantial overlap in perplexity between correct and incorrect rollouts.

**Taxonomy-Enhanced Scientific Retrieval** | *Group Lead* *01/2025 - 05/2025*
CS497 Team Project -- Advised by Prof. Jiawei Han [Tech Report]

- Investigated taxonomy-enhanced retrieval methods for scientific papers using LitSearch dataset (64K docs).
- Developed and evaluated multiple retrieval strategies—including taxonomy alignment, fusion embedding, and a fine-tuned encoder—demonstrating an improvement in Recall@5 from 39.0% to 47.5% (+8.5%).

## Technical Skills

**Languages:** Python | C/C++
**Core Frameworks & Tools:** PyTorch | vLLM | veRL | LlamaFactory | Slurm
**Systems & Training:** ZeRO/DeepSpeed | Retrieval-Augmented Generation | RLHF/RLVR
**Others:** Docker | MySQL | MongoDB | Kubernetes | FastAPI | Milvus | Flask | Neo4j

## Activities & Honors

- **Hackathons** – (2024 & 2025)
- **Datathons** – (2024 & 2025)
- **Course Assistant** – (08/2023 - 12/2023)
- **Dean's List** – (Spring 2025, Fall 2024, Spring 2023)

Last updated: Oct 13, 2026